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School of Computer Science | Final Year Project



DAILYKHATA

Deliverable 1: Target Market & Audience Analysis

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1. Introduction

DailyKhata is a personal expense management application primarily designed for Pakistani users who handle paper receipts, cash purchases and informal expense records. It aims to reduce the effort of recording spending and make those records easier to review. The proposed application combines receipt digitisation, quick expense entry, understandable categories, spending summaries, product price history and an AI Financial Assistant.

This deliverable identifies the intended audience, explains their problems and expectations, and establishes the high-level scope. Deliverable 2 examines the reported research and stakeholder needs. Deliverable 3 translates those findings and the agreed project decisions into requirements. The focus here is the value of the application to its users, rather than its technical implementation. [1]

DailyKhata is a mobile-first Final Year Project for a four-student BS Computer Science team. It uses Pakistani rupees (PKR) and targets ordinary Android-class devices. Its purpose is to help an account holder maintain personal or household expense records. It does not process payments or manage the user's bank account.

2. Market and User Context

The project belongs to financial technology, specifically personal expense management. Its target market is Pakistani smartphone users who want a practical record of everyday spending. The selected segment includes adults managing household purchases, student expenses or spending from a salary. The earlier project analysis uses an indicative age range of 18 to 50; this describes a design focus, not an eligibility limit or a measured market distribution. [2]

The relevant purchases do not all produce the same kind of information. A grocery receipt may contain several products and expense categories, while a rickshaw fare may have no receipt. A diary or phone note can preserve an amount, but the user must still enter it, remember its purpose and combine it with other entries during review. A useful solution therefore needs both receipt capture and a short route for purchases without receipts.

The reported project research consists of three interviews, informal observation and an examination of existing applications. It highlights repetitive entry, delayed recording and concern about incorrect categories. These qualitative findings support the proposed workflow, but do not establish market size, widespread adoption or the superiority of DailyKhata over current products. Detailed evidence and limitations appear in Deliverable 2. [3]

Local relevance comes from familiar spending labels, PKR amounts and attention to cash purchases. English and Urdu localisation is intended but deferrable. The core application must remain useful with an initial English interface and locally understandable categories. Language support should improve access when implemented, without becoming a condition for completing the main expense workflow.

3. Target Users

The primary users are homemakers and household budget managers, university students, and salaried professionals. Household budget managers need to organise everyday purchases and understand changes in grocery and household spending. Students need to record frequent small expenses with little effort. Salaried professionals need a coherent history that supports periodic review of monthly expenditure.

Small shopkeepers are secondary direct users. They may use the same expense workflow for petty cash, supplier expenses, receipt capture and summaries. This limited use does not introduce sales management, inventory, point-of-sale functionality or complete small-business accounting.

Family members and advisers are secondary beneficiaries of information shared by the account holder. A family member can review household totals with that person. An adviser can interpret information the user chooses to share. Neither role implies a separate portal, privileged access or collaborative editing.

DailyKhata uses a single-account model: one account holder records and manages their expenses, including household purchases. Shared household accounts, invitations and household permissions are future work. Digital confidence varies across all user groups, so the interface should support understandable review and correction without assuming accounting knowledge or advanced technical ability.

4. User Personas

The five personas below are design profiles used to explain needs and guide later evaluation. They are not five verified interview participants. Their ages and roles follow the agreed project definitions. The separate discrepancy in Usman's reported interview age is discussed in Deliverable 2. [2, 3]

Persona	Age and role	Goals, behaviour and expectations
Ayesha Tariq	34; homemaker and household budget manager; primary user	Handles receipts and cash purchases. Wants weekly household visibility and grocery price comparisons. Prefers photographing a receipt to transcription, with readable review, familiar categories and private records.
Bilal Ahmed	27; salaried professional; primary user	Reviews personal spending periodically and is comfortable with digital tools. Wants searchable monthly records and plain-language answers with minimal upkeep.
Hamza Sheikh	21; university student; primary user	Makes frequent small cash purchases and dislikes repeated forms. Wants quick text entry and useful category totals to understand personal spending and allowance use.

Persona	Age and role	Goals, behaviour and expectations
Zubair Hassan	Age not recorded; family member and spouse of a primary user; secondary beneficiary	Occasionally reviews spending with the account holder. Wants understandable household summaries, without requiring a separate household login.
Usman Raza	41; small shopkeeper; secondary direct user	Needs dependable petty-cash and supplier-expense records. Uses the same capture and summary workflow as other account holders, without sales, inventory or point-of-sale tools.

These profiles show why one flexible expense workflow is appropriate. Ayesha may start from a paper receipt, Hamza from a short cash description and Bilal from a monthly question. Each still needs a correctable record and a clear spending summary. Zubair benefits from understandable output, while Usman reuses the expense functions within a narrower secondary use case.

5. User Problems

Fragmented records. Paper receipts may be misplaced or become difficult to read. Notes, diaries and remembered amounts remain separate until the user assembles them. Finding a specific purchase or reconstructing a month can require reviewing many individual entries.

Unrecorded cash purchases. Small purchases without receipts depend on immediate entry or memory. Delayed recording can lose the date, purpose or precise amount. A receipt-only application would leave this part of the problem unresolved.

Repeated entry and classification. Transcribing receipt details and completing multiple fields can discourage continued use. A mixed receipt also requires more than one spending category. Automatic suggestions can reduce this work, but incorrect values must be easy to identify and correct.

Limited spending visibility. A list of notes does not itself explain which categories account for spending or how two periods differ. Users need consistent totals and a convenient way to ask questions without manually combining records.

Difficult product comparisons. A total for groceries does not reveal how the price of a repeatedly purchased item has changed. Useful comparisons need product identity, purchase date and quantity context. Different brands or pack sizes cannot automatically be treated as equivalent purchases.

Trust and privacy. Incorrect categories, duplicated amounts or inconsistent summaries weaken confidence. Financial records are also personal information. Users need control over corrections and access, as well as understandable explanations of automated output.

These needs connect convenience with dependable record keeping rather than treating speed as the only objective.

6. User Needs and Expectations

The proposed benefits follow directly from the recording and review problems. They are intended outcomes to evaluate, rather than improvements already demonstrated by testing.

User need	Expected response from DailyKhata
Capture purchases with less transcription	Photograph or upload a receipt, then review extracted information.
Record spending without a receipt	Use a short manual form or a text-based cash description, including past expenses.
Retain control over automation	Confirm or correct extracted values, categories and product interpretations before saving.
Understand recorded spending	Search history, inspect category totals and review monthly summaries.
Compare earlier purchases	View item-level prices with dates, product identity and available quantity information.
Obtain understandable answers	Ask the specialised assistant questions about the user's own records.
Keep dependable private records	Use authenticated access, protected storage and consistent editing and deletion.

Users should be able to see whether an entry is still being processed, needs correction or has been saved. A failed receipt extraction should offer retry or manual completion. After saving, the user should be able to find the expense again and correct or delete it. Those changes must also affect related summaries and price history.

Convenient interaction is important, but the project has not established a universal capture time. A short cash entry and a complex receipt require different amounts of work. Later evaluation should assess entry effort, processing delays and correction separately. Similarly, a spending summary represents recorded expenses; it cannot establish the user's complete spending when purchases are missing.

7. Proposed DailyKhata Solution

DailyKhata connects capture, review and later understanding in one expense ledger. A photographed or uploaded receipt is processed using optical character recognition (OCR). The application proposes structured information, including merchant and date where available, line items, available quantities, prices and the receipt total. The user reviews

and corrects the result before saving. Manual quick entry and text-based cash entry provide the core alternatives for purchases without receipts.

Automatic categorisation proposes categories for both transactions and individual receipt items. Familiar labels, such as Ration, help users recognise the purpose of spending. Behind these labels, a full internal Chart of Accounts and category-to-account mappings support accounting-compatible expense records. Users are not expected to enter account codes or journals. Wherever double-entry journals are used, they must balance; their implementation remains a design decision. This accounting structure does not turn DailyKhata into full SME accounting software.

Saved expenses support history, search, filtering, category totals, monthly summaries and a dashboard. Product information also supports item-level price history. Semantic product classification groups related products while retaining their distinct identities. For example, two detergent brands may share the product type Laundry Detergent, while their individual prices and pack sizes remain separate. Comparisons must explain missing quantities rather than imply equivalent value.

The AI Financial Assistant is specialised in the current user's DailyKhata records. It supports natural-language spending questions, summaries, period comparisons, recurring spending analysis, semantic product comparisons, historical price comparisons, unusually high recorded spending and personalised saving suggestions based on recorded history. It should distinguish recorded facts, calculations and suggestions, and explain when information is insufficient.

The assistant is read-only and cannot change expenses or perform financial actions. It is not an unrestricted general financial chatbot and does not provide investment, tax or banking advice, or unrelated financial education. Analysis of recurring past spending is included; automatically creating recurring expenses and predictive anomaly detection are future work.

8. High-Level Scope

Must Have means required for the minimum successful FYP. Should Have means intended but deferrable. Could Have means optional if time permits. Future Scope identifies possible later work, while Out of Scope identifies explicit exclusions from current delivery. These labels describe commitments, not implementation progress.

Priority	Capabilities
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Priority	Capabilities
Must Have	Registration and authentication; basic profile and settings; secure personal expense ledger; receipt photograph/upload, OCR, structured extraction and review; manual quick entry and text-based cash entry; transaction and line-item categorisation with confirmation and local labels; full internal Chart of Accounts, category mappings and accounting-compatible records; history, search, filtering, editing and deletion; category totals, monthly summaries and dashboard; semantic product classification and item-level price history; natural-language querying and the specialised AI Financial Assistant; reasonable support for ordinary Android-class devices.
Should Have	Voice-based cash entry; English and Urdu localisation; CSV and/or PDF export; custom labels for existing categories; simple spending visualisations; dedicated month-to-month comparison views; receipt image archive.
Could Have	User-created categories; offline draft expense entry; expense reminders; exportable household summaries.

The essential receipt output is a structured expense record. Retaining the receipt image beyond processing is Should Have, and permanent image retention is not promised. Local category labels are core even if bilingual localisation is deferred. Dedicated comparison screens are optional, while conversational comparisons through the assistant remain core. Offline drafts do not imply offline OCR or an offline assistant.

Current exclusions cover bank and wallet integration, automatic card synchronisation, open banking, payments and money transfers. They also cover sales and inventory workflows, point-of-sale, payroll, enterprise ERP, complete SME accounting, tax filing, investment management, credit scoring, fraud-detection infrastructure, commercial banking, enterprise organisation management, arbitrary financial integrations and banking-grade certification. Production infrastructure for millions of concurrent users is outside this FYP.

Possible future work includes bank importing, wallet and retailer integrations, e-receipt ingestion, shared household accounts, collaborative budgeting, business accounting, inventory and point-of-sale functions. Recurring expense creation, budgeting goals, predictive analytics, anomaly detection, advanced financial recommendations, deeper Urdu support, additional languages and multiple currencies are also future possibilities. Their inclusion here creates no current delivery commitment.

9. Target Market Fit

The selected audience shares an expense-recording problem despite differences in purchasing habits. Receipt capture addresses transcription, quick text entry addresses no-receipt purchases, and review protects users from incorrect automated results. Search, summaries and the assistant make the resulting record useful after capture. This

relationship between recording effort and later value provides the project's principal market rationale.

The design also avoids requiring users to adopt professional accounting practices. Familiar categories remain visible while structured accounts support consistency internally. A household account holder can understand spending without creating a multi-user household system. A small shopkeeper can record expenses without requiring the team to build a separate business application.

Feasibility depends on integrating a coherent core workflow before optional extensions. Existing OCR and language-model services may be used; training a foundation model is not a project commitment. Representative receipt and user evaluation should establish whether people can correct extraction, record cash purchases, retrieve transactions and understand assistant answers. The project does not yet claim improved retention, financial savings or verified market demand.

10. Conclusion

DailyKhata proposes a focused response to fragmented personal and household expense records. Its combination of receipt and cash capture, user-controlled correction, structured storage and understandable analysis fits the needs of the selected users. Deliverable 2 explains the reported evidence behind those needs, and Deliverable 3 defines the requirements for implementing and evaluating the proposed solution.